Clinical Psychology Review, Vol. 8, pp. 611–636, 1988
Printed in the USA. All rights reserved.

0272-7358/88 $3.00 + .00
Copyright © 1988 Pergamon Press plc

ATTACHMENT THEORY, PERSONALITY DEVELOPMENT, AND PSYCHOTHERAPY

Randolph J. Paterson
Greg Moran

University of Western Ontario

ABSTRACT. *Attachment theory, currently a dominant theme in the study of early social development, is beginning to have a presence in clinical psychology as well. A review of the theory as proposed by Bowlby is followed by an examination of the related research literature. Research using Ainsworth's Strange Situation paradigm suggests that the infant's attachment status is a genuine product of the interaction between mother and child rather than a measure of infant temperament, and is related to a broad array of developmental variables. Relatively little research has effectively examined the link between the mother-infant relationship and adult social functioning, however, due in part to difficulties with the measurement of attachment-related constructs. It is concluded that while the theory continues to evolve and some critical issues remain unresolved, attachment may be a useful construct for conceptualizing many disorders, and has the potential to provide valuable insights regarding the process and techniques of psychotherapy.*

In recent years attachment theory has been applied with growing frequency to the practice of psychotherapy, particularly by those who favor a psychodynamic approach (e.g., Turco, 1982). In its ethological origins and expanding research base, attachment theory seems to provide a welcome empirical justification for a central tenet of psychodynamic theory: That early life experiences strongly influence subsequent adult functioning and vulnerability to psychopathology. The empirical literature on attachment is immense; it is presently a major force in the study of early social development. But practitioners not overly familiar with the recent developmental literature may find themselves somewhat at sea when attempting to evaluate this new influence in the clinical setting.

The present article is divided into two main sections in order to address the issues of theory and research. First, a brief review of the history and nature of attachment theory is presented. Primary emphasis is given to the writings of John Bowlby. In the second and larger section the empirical literature is examined. An

· for reprints should be sent to Randolph J. Paterson, Department of Psychology, ᶠ Western Ontario, London, Ontario, Canada N6A 5C2.

exhaustive review is beyond the scope of this article or, for that matter, any presentation short of a multi-volume work. Instead, the focus is limited to issues that have provided the primary impetus for research or have strong implications for clinicians. Although at present the research literature offers the clearest guidance to those working with children, such readers may be more familiar with the attachment literature than specialists in adult work and may wish to pursue these issues in more depth than can be presented here. Consequently, greater emphasis is given to issues dealing with adulthood than is often the case in discussions of attachment theory.

THE ORIGINS AND PRINCIPLES OF ATTACHMENT THEORY

The origins of attachment theory can be traced to John Bowlby, a psychiatrist whose work in residential facilities for children and adolescents centered on the role of environmental factors in development. Trained in the psychoanalytic tradition, he became convinced of the paramount importance of the relationship in the first years of life between the infant and the primary caretaker in the development of the self and of later social behavior (Bowlby, 1958). Bowlby joined with the learning theorists of the time in criticizing psychoanalytic theory's inadequate empirical basis, but he also appreciated the difficulty inherent in testing complex psychoanalytic concepts. In his search for a way out of this conundrum, he turned to ethology.

Bowlby (1957) found that ethologists had discarded simple catalogs of "instincts" as being oversimplifications of complex behavioral processes which are dependent upon extensive environmental feedback and serve multiple goals. They had progressed to thorough descriptions of these behavior systems and the analysis of their adaptive function and evolutionary origins. Psychoanalytic theory shared ethology's early reliance on the concept of instinct. Bowlby concluded that a more rigorous description of human behavior and consideration of its adaptive function would serve psychoanalysis as it had ethology. His subsequent work drew heavily on ethological concepts, culminating in the publication of his influential trilogy, *Attachment and Loss* (1969, 1973, 1980).

Two animal research programs are of particular interest in association with attachment theory. Lorenz' (1952) studies of imprinting demonstrated the tendency of the young of some species to bond to a specific individual soon after birth. Related work by Harlow and his colleagues extended these findings to the attachment behavior of monkeys. The best known of these studies demonstrated that, given a choice, infant rhesus monkeys spend more time with a non-lactating cloth-covered surrogate mother than with a lactating one made of wire (Harlow, 1958). This finding indicates that the infant monkey's attachment to its mother does not rest solely on her status as a secondary reinforcer based on an association with food. It has become clear that infants of many species are biologically prepared to develop an attachment to the mother or a mother-figure independent of the direct reinforcement of known physiological needs. Mothers are likewise prepared to respond in specific ways to the presence or absence of their infants, who are equipped with behavior patterns (e.g., crying) designed to elicit caring responses (Harlow, 1971; Jensen, 1965).

Bowlby (1957) interpreted these data in terms of their adaptive significance. Infant monkeys are relatively helpless (and infant humans even more so), yet they

have a wide variety of needs. They must have nourishment, water, protection from predators, and so on. Without the support of others the infant would perish. It is adaptive for the infant to develop an attachment to another individual who will satisfy these needs until the infant can meet them him/herself. So too, the reproductive success of the mother is enhanced if she helps the infant. The stage is set for the development of a behavior system in which the mother is biologically "ready" (or prepared to learn) to perform specific caring actions and the infant is ready to perform actions that will elicit such caregiving. Harlow (1971) and Bowlby (1984) have both presented accounts of this system, and where they cover the same ground they are largely in agreement. But it is Bowlby whose work has provided the basis for much subsequent research with humans, and his influence that is at the core of attachment theory today.

Bowlby defines an attachment as a bond developed with "some other differentiated and preferred individual, who is usually conceived as stronger and/or wiser" (1977a, p. 203). Attachment behavior is "any form of behavior that results in a person attaining or retaining proximity" (1977a, p. 203) to an attachment figure. Bowlby (1977a) has listed the defining features of attachments and attachment behavior: Attachments are directed toward specific individuals, are characterized by their long duration, and serve the biological function of survival. They have a strongly positive emotional component while they are being formed (falling in love), maintained (security), or renewed (joy), and a strongly negative one when they are threatened (anxiety and/or anger) or lost (sorrow and grieving). Since attachment is a primary and not a secondary drive, attachments may develop independently of direct reinforcement by the attachment figure (as was the case for Harlow's monkeys). Finally, attachment behavior is controlled by a specific set of conditions for activation (such as fear) and termination (such as interaction with the attachment figure).

The theory clearly states that the attachment system is a central feature of the developing relationship between infants and their caretakers. Bowlby further suggests that attachment plays a role beyond infancy. He states that attachment behavior is most easily activated before the age of three, and with declining ease thereafter (1984). But, while an adult is clearly more self-sufficient than an infant and not in need of constant attention from a caregiver, there are certain situations in which a more knowledgeable or experienced individual is of great help. The range of situations in which attachment behavior is activated may become more narrow, but will not vanish altogether. An attachment figure may be perceived as stronger and/or wiser in only some situations, thus allowing for attachment to occur among relative equals, as in strong friendships and romantic relationships (1977a).

Attachment may also affect adult functioning in a more significant way. Through repeated transactions with its attachment figures, the infant is said to form internal representations both of the self and of others. These "internal working models" (Bowlby, 1973) subsequently form a heuristic basis for future relationships. The model of the self structures expectations about one's own role in relationships. For example, an infant whose needs are typically left unmet may develop a model of the self as being unworthy of the attention of others and incapable of influencing their behavior. The model for others serves as a template for future relationships, forming the fundamental assumptions and beliefs about how others will behave. The adult neglected by mother in infancy may believe that

each new person met will prove to be inaccessible, uncaring, and unresponsive. Bowlby (1980) suggests that these models act outside conscious awareness and so are resistant to change. They are most relevant in close relationships involving attachment, and relationships simulating the caretaker–recipient bond (such as relations with authority figures), but they may also affect relations with friends and acquaintances.

Some attachment behaviors, particularly the more passive varieties such as crying, are present from birth. Others, notably the "social smile," appear over the ensuing weeks and months. Initially the attachment "relationship" is somewhat one-sided, as at birth the infant is incapable of discriminating the primary caretaker (usually mother, and who will be referred to as such henceforth) from others and so does not develop an exclusive attachment relationship with her immediately. From an ethological perspective this is nonessential, as the infant is not yet mobile. At this stage attachment behavior may instead encourage the development of the mother's bond to the infant, thereby promoting her proximity and caregiving.

By about 6 months, however, the infant recognizes and has developed an exclusive relationship with the mother. This is now an adaptive feature, since new behaviors such as crawling, walking, and speech begin to enter the repertoire and can be used in the service of attachment if the infant can recognize a particular individual to whom he or she can approach and call. Between 6 months and 1 year the infant commonly develops "stranger anxiety"; unfamiliar figures are met with resistance and anxiety. While this typically appears soon after recognition of the mother as a separate person, it is not considered an index of the quality of the attachment with her. Instead it seems to serve the adaptive function of encouraging the infant to avoid potentially dangerous, noncaregiving others in favor of the familiar caregiver. Over subsequent months and years stranger anxiety diminishes. Others frequently in the presence of the child may also become attachment figures, but the mother usually remains primary (Bowlby, 1977a).

Once an attachment is established, the infant uses the mother as a secure base from which to explore the environment. Ainsworth (1982) emphasizes the importance of the exploratory behavior system which, like the attachment behavior system, is essential to the individual's future welfare. When the child "is rested, healthy, and not hungry, his attachment system is likely to be inactive" (Ainsworth, 1982, p. 4). Rather than exhibiting attachment behavior designed to increase proximity to mother, the exploratory system is activated and the infant (utilizing many of the same skills employed by the attachment system) moves away to examine and manipulate the surroundings. In this way skills in dealing with the environment are developed.

As time passes, the distance from mother increases, or an apparent danger arises, the attachment system is reactivated and the infant attempts to reestablish contact by moving toward the mother, by emitting attachment behavior (calling or crying) designed to bring her, or simply by some form of social referencing such as establishing eye contact. Once the presence and availability of the secure base has been established, the infant moves away again and recommences exploration. The exploratory system is most likely to be activated by novelty, and the attachment system by threat. The more strongly activated system gains expression (Ainsworth, 1982). These complementary, mutually inhibitory systems ensure

that the infant learns about the environment but does not become lost or exposed to danger unnecessarily.

The theory's claim that exploratory behavior occurs more often when the secure base is available than when it is not raises an important point. Rather than conceptualizing dependence on mother and the development of autonomy as being antagonistic dynamics, the two are seen as mutually facilitative. Harlow (1971) found that infant monkeys placed in a strange test room would first cling to their surrogate cloth mothers, then explore their surroundings. In contrast, those placed alone in the room froze and did not explore. Although attachment and exploratory behaviors are incompatible, the attachment figure must be available for the exploratory system to find outlet.

It follows that the unwilling separation of the young child from the mother should activate the attachment system. Based on both casual observation and studies in residential nurseries and hospitals, Bowlby (1958, 1961) has described a consistent behavioral sequence following such separations. The initial phase is characterized by crying and demands for mother's return that potentially function to draw her attention or the attention of others who will bring her. If this fails, then despair sets in, characterized by continued longing for mother but without hope that she will actually return. The attendant depression and inactivity may give way, on occasion, to outbursts of rage. Finally, the child becomes detached from the problem, appearing to have forgotten the mother. When she eventually returns, the child may initially be unresponsive or avoidant of the mother. Provided the separation has not been prolonged (such as over 6 months), this avoidance gives way to intense clinging, anxiety, and anger. Exploratory behavior is sharply curtailed, as though the child does not trust the mother to remain available. Over time the relationship may gradually return to normal.

Inadequacy of the attachment figure may be expressed in ways other than physical absence. The primary function of a parent, according to Bowlby, is to provide a secure base and encourage exploration from it (1977a). The parent must be readily available to the child, responsive to its needs, and accepting of the child as a person. The child should not be subjected to threats of withdrawal of love or caring, or threats to abandon the family. Finally, the child should not be made to feel responsible for the parent's illnesses or death. All of these interfere with the child's perception of the parent as a secure base and may, Bowlby suggests, have far-reaching consequences. If the course of normal attachment is disrupted in these ways or by traumatic separation (either repeated, extended, or permanent), then the child can be considered at risk to develop a generalized fear of losing its attachment figures.

Several patterns of disorder can develop from this anxiety. Bowlby (1977a) suggests that the most common is that of "anxious attachment," in which the individual is constantly fearful of losing attachment figures with the result that attachment behavior is activated too easily. According to Bretherton (1985), this pattern may result when the child "has to take more than optimal responsibility for maintaining proximity" (p. 11). The pattern is analogous to the clinging response demonstrated toward parents upon reunion after a separation. The anxiously attached individual is said to become excessively clinging and jealous in relationships, and often attempts to test and verify the strength of bonds by being manipulative. Of course, such a pattern tends to disrupt attachments and so may

become self-perpetuating. Such individuals, continually fearing desertion, are liable to harbour high levels of anger toward their attachment figures. At the same time, they are likely to be unable to express anger effectively because they fear this could lead to the feared event itself—loss of the attachment. In combination, these features result in the displacement of much anger onto others. The pattern is reminiscent of the DSM-III-R (American Psychiatric Association, 1987) diagnoses of borderline and dependent personality disorders.

If anxious attachment is similar to the clinging stage of childhood reunion, then "compulsive self-reliance" (Bowlby, 1977a) is parallel to the more pathological withdrawal period. Compulsively self-reliant individuals have essentially lost hope of finding an adequate attachment figure, and avoid close relationships. Their apparent independence is largely illusory, motivated not by free choice or an urge to explore and develop their own resources, but by a fear of attachments and their anticipated consequences. The absence of attachments in adulthood may suppress growth-oriented exploratory behavior in these individuals, just as absence of the attachment figure does in childhood. The description of these individuals suggests the DSM-III-R diagnoses of schizoid and avoidant personality disorders.

Bowlby suggests other patterns of disorder which find their origins in disturbed attachment histories, but they are variations on these two themes. For example, compulsive caregiving (1977a) refers to an individual who denies personal needs to fulfill those of others. The endlessly self-sacrificing mother (the "don't-worry-about-me" syndrome) is a salient example. Bowlby sees this pattern in part as compulsive self-reliance, but it may also be a manipulative strategy to elicit gratitude in the form of a sustained relationship—a pattern related more to anxious attachment. Each of the problems can arise if the base is not secure or if the parent fails to encourage the child to explore. The specific form of the attachment deficits will determine the pattern that arises. However, Bowlby does not fully elaborate on this mechanism or the specific links between early attachment and adult disorder.

Early attachment is most clearly disrupted when no primary attachment figure is available, or when attachments are formed only to have the attachment figure become permanently unavailable. Consequently, much of the clinically oriented literature on attachment has focussed on the results of separation from and loss of attachment figures. Such loss is more easily identified than less understood or quantifiable aberrations of the early attachment relationship. Bereavement thus provides an excellent illustration of the application of attachment theory to human functioning and psychopathology.

The normal course of mourning a lost attachment has been described by Bowlby (1961; Bowlby & Parkes, 1970) and by Parkes (1972). Bowlby and Parkes (1970) describe four stages, of which the latter three closely parallel the stages of the child's reaction to separation. The first and usually shortest stage (lasting from several hours to several days) is characterized by numbing and a lack of affect, occasionally broken by sudden displays of anger or anxiety. The second stage features intense longing for the lost figure and can often last for several months. Parkes (1972) suggests that the representation of the world at this stage continues to include the presence of the lost figure. The individual automatically interprets various stimuli within that context. A face on the street may cause momentary "recognition" of the deceased, or an extra place may be set for a meal. Alarm and

searching responses are manifestations of attachment behavior designed to regain the lost figure. In situations of temporary loss such behavior would be adaptive. As in anxious attachment, this stage of loss parallels descriptions of the child's initial response to separation.

Eventually the world view in which the lost figure is alive breaks down and is replaced by disorganization and despair, the third stage (Bowlby & Parkes, 1970). Feelings of emptiness are common, perhaps because interaction patterns associated with the figure are not activated and have not been replaced by other activities. Anger and other emotions become milder, and pangs of grief are replaced by longer periods of apathy and depression (Parkes, 1972). This stage resembles the withdrawal reported in longer childhood separations. In the final stage, the individual reintegrates the world view into a new model excluding the lost figure. New roles are adopted, and new behaviors not dependent on the other's presence are produced. Here again, this pattern reflects the final stage of childhood separation, when the attachment is broken and attachments with others may be formed in its place.

The process of bereavement is complicated by a number of variables that may lead to significant problems of adjustment. Maladaptive patterns of attachment, such as anxious attachment or compulsive self-reliance may arise particularly if the loss occurs at an early age. Bowlby and Parkes (1970) argue that grieving a death is more difficult for children than adults. Children seem to find it difficult to understand the finality of death, and their extreme dependence imparts a much greater threat to the loss of the attachment. In addition, the strength of childhood attachments (Bowlby, 1977a) predisposes them to more difficulty when a central attachment figure dies. This can only be exacerbated by the frequently inadequate understanding of the child's position by surviving adults. Raphael (1982), in a study of families in which one parent had died, found that many children had difficulty understanding the concept of death. Many more were not taken to the funeral, and some had not even been told of the death. The surviving parent often believed that the child had recovered quickly, attributing behavioral disturbances to other causes, or missing such problems completely in the haze of their own loss. It seems that while children are often the most affected in these situations, they frequently have the least support. If loss can lead to subsequent disorder, as attachment theory suggests, then loss in childhood is likely to be particularly damaging.

Both adults and children sometimes avoid grief altogether, or fail to progress through the stages of the grief process (Parkes, 1972). Both patterns can lead to unusually prolonged disturbances of functioning. Several factors can predispose the individual to a pathological grief reaction (Parkes, 1972). These include a pre-existing problem with attachments, such as the excessive dependency of the anxiously attached individual. The experience of loss could make an anxiously attached person even more conscious of the precariousness of attachments, and the compulsively self-reliant individual yet more reluctant to engage in them. Another predictor of problems in grieving is pre-existing ambivalence toward the lost figure, such that the expression of emotions essential to successful grieving would necessitate the recognition of hostile affect just when such bond-threatening feelings are likely to be avoided most.

While the above discussion is directed specifically at instances of bereavement,

analogous processes are postulated for other types of loss. To the extent that the primary attachment figure fails to provide a secure base, inadequate caretaking is also a situation of deprivation. Attachment theory thus attempts to relate early interactions with caretakers to later development, and to certain patterns of pathology (Bowlby, 1977b). Given that most psychopathology involves a social component, attachment theory could have at least indirect relevance to most clinical cases.

RESEARCH ON ATTACHMENT

Failure to Thrive

One of the more basic principles of attachment theory is that the infant has a primary need to develop a relationship with a caretaker. Harlow's early studies (1958) led him to assert an inborn desire for contact comfort in rhesus monkey infants. If the need is indeed primary, then failure to form an attachment should produce adverse consequences, even when other physiological needs are met. Such is the situation of some children who have either been neglected by their parents (Pollitt, Eichler, & Chan, 1975), or who have been raised in institutional settings where individualized attention was not provided (Provence & Lipton, 1962; Spitz, 1945). These infants may exhibit normal attachment behaviors, such as crying, smiling, and babbling, up to 6 months of age. But in the absence of an appropriate object for their attachment, the behavior patterns disappear. At 1 year these children rarely vocalize and become rigid when picked up. They also typically display restricted growth, delayed developmental milestones (including speech), poor motor control, greater vulnerability to disease, and general weakness (Tibbits-Kleber & Howell, 1985). Disturbances of feeding and digestion are common (Call, 1984), and social responsiveness is very poor (Tibbits-Kleber & Howell, 1985).

This cluster of symptoms attendant upon failure to develop attachments is now well accepted. The associated diagnostic category, Reactive Attachment Disorder of Infancy (RAD), is attributed in DSM-III-R to a lack of adequate caretaking (American Psychiatric Association, 1987). There is disagreement over whether RAD is directly attributable to inadequate stimulation or more specifically to the absence of a unique attachment to a single caretaker. Provence and Lipton (1962) have shown that institutionalized infants given toys and other sensory stimulation but only limited attention by caregivers were substantially as impaired as infants not given such stimulation. Further, several studies have indicated that individualized caretaking can overcome the symptoms of RAD, provided that such care is initiated before the age of 3 years (Clark & Hanisee, 1982; Dennis, 1973; Rutter, 1979). Attachment itself, rather than stimulation, appears to be critical. Some studies have found that infants given adequate care by multiple caregivers seem perfectly normal (Levy-Shiff, 1983). This suggests that a unique relationship may not be essential, although some infants raised with multiple caregivers are found to develop specialized attachments to one or two of them (e.g., Marvin, 1977).

In sum, there is substantial evidence that simply satisfying the physiological needs of infants in the absence of adequate social stimulation is insufficient. The specific requirements of relationships with caregivers are not well defined. However, it is clear that the absence of such relationships has a significant detrimental impact on many areas of early development.

Maternal Bonding

The past 15 years have witnessed the popularization of the notion that the period immediately after birth is critical to the development of the mother's bond to her infant. The theory is sometimes likened to imprinting—mistakenly so, since imprinting involves the newborn fixating on a being it identifies as "mother" independently of what the mother may feel about it. Such bonding is useful in precocial animals, as it prevents mobile young from straying too far from their caretaker, but is less important in altricial species such as humans. Since human babies are not capable of discriminating one individual from another until several months after birth, any bonding that occurs at this stage must be from mother to infant.

The most influential proponents of the critical period hypothesis have been Marshall Klaus and John Kennell. In a longitudinal study, control mothers were shown their babies briefly following birth and had only short visits for the next 3 days, while experimental group mothers were given 1 hour of contact within 2 hours of birth and spent extra time with them for the next 3 days. At 1 month and 1 year, several measures indicated that the experimental mothers were more closely involved with their babies (Klaus et al., 1972; Kennell et al., 1974), and at 1 year experimental group babies scored significantly higher on the Bayley developmental assessment. At 2 years, mothers in the experimental group used more elaborate and varied child-directed speech than controls (Ringler, Kennell, Jarvella, Navojosky, & Klaus, 1975). At age 5 the groups did not differ in intelligence (Ringler, Trause, Klaus, & Kennell, 1978).

Klaus and Kennell (1976, 1982) have interpreted the results to be supportive of the critical period hypothesis. However, subsequent reviews (Lamb, 1982; Myers, 1984) have pointed out a number of methodological problems, including a failure to adequately account for Type I errors in view of the large number of variables examined, a trend toward the decay of effects over time, and the failure to replicate findings across studies.

A number of recent studies have further undermined the hypothesis. Gomes-Pedro, de Almeida, da Costa, and Barbosa (1984) found that initial differences favoring an early-contact group tended to decline over time, and Svejda, Campos, and Emde (1980) found no differences on measures of mother–infant interaction only 36 hours after delivery. A supposed consequence of the failure of bond formation is that the mother will be less motivated to take good care of her child. But Egeland and Vaughn (1981) found that delivery complications, prematurity, hospitalization time, and other indications of potential bonding failure all failed to distinguish between mothers identified as providing suboptimal versus superior care.

In her review of the literature, Myers (1984) points out that if the critical period is indeed critical, then experiences during that period should have effects that amplify as the child develops, "fanning out" across measures and traits. In contrast, any gains attributable to early mother–infant contact seem to disappear over time. It would appear that the period soon after birth may simply be a relatively sensitive (rather than critical) period, perhaps due to a primacy effect or the ordeal of birth. To the degree that the critical period hypothesis has encouraged hospital maternity wards to allow early contact and humanize the birthing process, it is to be applauded. There is, however, little evidence that early contact plays a critical role in bond formation.

Ainsworth's Strange Situation

Infants are presumed to use a variety of behaviors in the service of attachment, including crying, smiling, establishing eye contact with the attachment figure, and crawling or walking toward her (Bowlby, 1969). In past years several studies have assessed the practicality of using these behaviors as indicators of individual differences in attachment status (e.g., Coates, Anderson, & Hartup, 1972). Unfortunately, disappointingly low correlations between these supposedly related measures were typically obtained. Lamb (1974) interpreted these results by observing that different behaviors are called into play in different situations, each acting independently in the service of the same goal: increasing proximity to the mother. An infant seeking to increase proximity would not cry, smile, and run to mother all at the same time, but would select whichever behavior was perceived to be most appropriate and most likely to succeed under the circumstances at hand. Given this, low correlations between the different behaviors would be expected. According to this view, differences in attachment would have to be assessed according to overall patterns of behavior in carefully delimited, replicable circumstances.

Enter Ainsworth and Wittig's (1969) longitudinal study of attachment in a Baltimore sample. They developed a scenario, the Strange Situation, designed to allow the assessment of the 12- to 24-month-old infant's relationship to a caretaker in a series of circumstances:

> An infant and mother are brought into a comfortable laboratory room; a stranger enters and sits talking to the mother and then to the infant; the mother leaves the room unobtrusively; the mother returns and the stranger leaves them together; the mother leaves the infant alone in the room; the stranger returns; the mother returns once more. Each of these seven episodes lasts three minutes unless the infant is more than mildly distressed. (Ainsworth, 1982, pp. 6–7)

The episodes typically become increasingly stressful for the infant, thus serving to elicit attachment behaviors. The procedure provides a standardized context within which attachment and exploratory behavior systems can be assessed. The session is videotaped and is later scored on scales of proximity and contact seeking, contact maintaining, resistance, avoidance, search, and interaction at a distance. The infant's responses to reunions with mother are frequently regarded as the most telling clues to attachment status.

Infants are classified into one of three groups. Infants classified as type B are said to be securely attached, using the mother as a secure base, perhaps becoming upset at separation, and greeting her upon reunion, returning to exploration afterward. Those classified as A or C are said to be anxiously attached. A's are considered avoidant: they substantially ignore the caretaker when playing and ignore or avoid her when she returns. There is little proximity-seeking. This seems to resemble an early form of compulsive self-reliance as described by Bowlby (1977a). Type C's are called resistant or ambivalent, mixing contact-seeking with angry rejection of the caretaker, and are so focused on the caretaker that they do not play independently. This is reminiscent of Bowlby's description of anxious attachment in the adult. In addition, there are eight attachment subclassifications, but the majority of studies use only the more general categories.

More recently, some researchers have proposed the use of a fourth category that soon may have an important impact on research in this area: the insecure-disorganized/disoriented or D infant (Main, Kaplan, & Cassidy, 1985). These infants display considerable apparent confusion and behavioral disorganization upon reunion with the parent, including undirected affect, contradictory patterns and incomplete movements. This category is only now being employed in studies of attachment and therefore does not figure in this review.

Many researchers collapse categories A and C to form a dichotomous classification of securely versus anxiously attached infants. In their review Lamb, Thompson, Gardner, Charnov, and Estes (1984) give some support to this practice, suggesting that there is more evidence supporting the distinction between B and non-B infants than between A and C infants. However, to the degree that A and C groups vary in underlying dynamics, the practice may have the disadvantage of introducing more variance than necessary to the insecure group, perhaps clouding potential relationships with other variables.

The controlled setting of the Strange Situation is an advantage over more naturalistic observation techniques, but prompts questions of external validity. However, several well-controlled studies have shown that behavior in the Strange Situation is reflective of the mother–infant relationship at home (Ainsworth, 1982; Gaensbauer et al., 1985).

Attachment, Infant Temperament, and Internal Models. A more serious debate in the literature has concerned whether infant behavior in the Strange Situation is truly attributable to the nature of the relationship with mother or simply reflects infant temperament. Kagan (1982) has taken the latter position, arguing that an infant's own predisposition to respond to desertion and reunion in specified ways, regardless of the status of the relationship with the caregiver, may be the primary determinant of attachment classifications received. Sroufe (1985) disagrees, arguing that the bulk of evidence supports the interpretation of Strange Situation behavior as a product of the caregiver–infant relationship. A number of findings which are of interest in their own right can be brought to bear on this issue.

Attachment Stability Over Time. One line of research relevant to this debate is the degree to which attachment status remains consistent over time. A widely accepted view states that if Strange Situation behavior reflects the relationship, then attachment status should remain stable in the absence of major stresses on the relationship or either partner, but should show less stability in the face of such stress. The majority of studies demonstrate high levels of stability over time. In a summary of nine studies that have examined this issue that was compiled by Lamb, Thompson, Gardner, & Charnov (1985), the average concordance (weighted for sample size) for middle-income families and families rated as providing excellent care was 77.1%.

Other studies have examined less stable families. Vaughn, Egeland, Sroufe, and Waters (1979) assessed attachment in low SES families at 12 and 18 months, finding only 62% concordance overall, and an association between stressful events acting on the family and shifts away from the optimal B classification to an insecure attachment. Egeland and Farber (1984) report a 60% 12- to 18-month concordance in a low SES sample. Egeland and Sroufe (1981a, 1981b) drew from

this pool two subsamples identified as providing seriously inadequate and excellent care. The concordance rate in the inadequate care group was 48%, compared to 81% for the excellent care group. Thompson, Lamb, and Estes (1982) found that maternal employment and the start of nonmaternal care can significantly shift attachment classification. Although some infants were observed to change from secure to insecure, others went from insecure to secure.

Overall, the data on temporal stability tend to support those arguing that attachment classification is a reflection of the interaction between mother and infant, rather than infant temperament alone. When family life is reasonably stable, attachment classifications also tend to be stable. In more stressful environments attachment status seems to change more often, and tends to change toward insecure attachment. This evidence is more difficult to explain from the temperament perspective.

Attachment Status Across Caretakers. Research examining the security of attachment status across different caretakers also supports the relationship view. If classification was a product of infant temperament, then one would expect the infant to achieve the same classification with mother and with father. Several studies have shown a moderate tendency for the attachment classifications achieved with different parents to be concordant at higher than chance levels (Belsky, Garduque, & Hrncir, 1984; Lamb, 1978; Lamb, Hwang, Frodi, & Frodi, 1982; Owen & Chase-Lansdale, 1982). However, the majority of studies find no significant relationship between the two classifications (Grossmann & Grossmann, 1981; Grossmann, Grossmann, Huber, & Wartner, 1981; Lamb, 1977; Main & Weston, 1981; Sagi et al., 1985). In fact, Main, Kaplan, and Cassidy (1985) report a zero correlation between attachment security toward mother and toward father.

Maternal Personality and Disturbance. Finally, some researchers have examined whether mothers' personality characteristics or caretaking styles can be reliably related to the attachment classifications achieved by their infants. If the attachment classification is the product of the relationship, one would expect it in turn to be related to maternal caretaking style and, perhaps more indirectly, to maternal personality characteristics. Tronick, Ricks and Cohn (1982) found that mothers of securely attached infants, as assessed at 12 months, rated themselves as higher in self-esteem, more likable, and more competent than mothers of insecurely attached infants. In a prospective study of maternal personality characteristics, Egeland and Farber (1984) found few differences between groups, although mothers of securely attached infants were found to be less aggressive than mothers of insecure infants. Benn (1986) has reported a significant association between a global measure of maternal emotional functioning and security of attachment, with better-functioning mothers having a higher proportion of securely attached children.

Several studies have looked more specifically at attachment to mothers suffering from psychological disorders. Radke-Yarrow, Cummings, Kuczynski, and Chapman (1985) found insecure attachment to be more common among children of mothers with unipolar and bipolar depression than among children of mildly depressed or normal mothers. Naslund, Persson-Blennow, McNeil, Kaij, and Malmquist-Larsson (1984), using an abbreviated version of the Strange Situa-

tion, found a significantly greater incidence of anxious attachment among the children of schizophrenic mothers relative to controls. D'Angelo (1986) found that the regular Strange Situation used with the same sample revealed a significant tendency for children of schizophrenics to be avoidant, and for children of depressed mothers to be ambivalently attached.

Caretaking Style and Attachment Status. The effects of personality and psychiatric disturbance, if casual, may be mediated by caretaking style. Ainsworth, Blehar, Waters, and Wall (1978) found that avoidant attachment at 1 year was associated with maternal aversion to physical contact, nonresponsiveness to signals such as crying, and a lack of emotional expressiveness. In contrast, ambivalent attachment was associated with inconsistency in responding to infants' signals, rather than simple low responding. Infants whose mothers responded to their cries in the first quarter cried less in the fourth quarter. This is strictly counter to what social learning theory would predict, since "reinforcement" of crying by responding reliably would be expected to lead to increased rates of crying later. Positive relationships between maternal responsivity and security of attachment have since been found by other researchers (Bates, Maslin, & Frankel, 1985; Belsky, Rovine, & Taylor, 1984), although Belsky et al. found no relation between security of attachment and frequency of close physical contact with their mothers. Egeland and Farber (1984) found that mothers of securely attached infants were more sensitive, perceptive, and empathetic than others.

Not surprisingly, outright abuse or neglect also relates to attachment security. Lamb, Gaensbauer, Malkin, and Schultz (1985) found that infants maltreated by their mothers were more likely to develop insecure attachments with them and with subsequent foster mothers, most often becoming avoidant; although children abused by others demonstrated no greater than usual tendency toward insecure attachment with their natural mothers. Others have also found a tendency for abuse or neglect to lead to insecure attachment (Schneider-Rosen & Cicchetti, 1984; Schneider-Rosen, Braunwald, Carlson, & Cicchetti, 1985). Lamb et al. (1985) have suggested that the overstimulation of chronic abuse may encourage the development of avoidant patterns, while the understimulation of neglect might lead to ambivalent attachment. Egeland and Sroufe (1981a), however, found that poor mothering was associated with different patterns over time. At 12 months such infants exhibited the ambivalent pattern, as though unsure how to respond to the parent; while at 18 months avoidant attachment was more common, as though by then they had given up attempting to relate.

Summary. These studies underline a relation between maternal characteristics and attachment classifications. Mothers of more securely attached infants appear to be more confident, emotionally stable, and responsive to their infants' attachment behaviors than mothers of insecurely attached infants. Futher, mothers suffering from mental illness and those who neglect or abuse their children tend to have insecurely attached infants. These patterns do not follow from the view that infant temperament is a major determinant of attachment. In summary, although infant temperament may have a role in attachment, the research on stability, concordance, and maternal influence suggests that attachment status primarily reflects the quality of the relationship between mother and child.

But while the accumulated evidence calls the temperament position into serious question, it also poses a problem for attachment theory. Evidence that attachment status changes over time and differs by attachment figure appears to conflict with the position that early attachment experience forms the basis of a unitary and stable internal working model that determines the form of subsequent relationships. While far from being resolved, some directions for approaching this issue are discernable from the existing literature. Multiple changing attachments might contribute to a single but very complex working model which is then applied universally. Alternatively, different attachment experiences might create a number of distinct working models, which might differ in their influence on subsequent relationships depending on the importance of the experience and the type of relationship involved.

Bretherton (1985) supports the view that the individual internalizes different working models corresponding to different formative relationships. As an illustration she offers Cottrell's (1969) case history of a man who seems to have based his harsh actions toward his son on his stormy experience with his father, and his loving relationship with his wife on his history with his mother. In a similar vein, psychodynamic clinicians often view the patient's behavior toward a male therapist to be reminiscent of the past relationship with father, and that toward a female therapist to be suggestive of the history with mother.

To summarize this position, the individual may internalize many early relationships; their influence on an adult relationship may depend on the similarity between the person concerned (or the position they hold) and the original prototype. In addition, stronger attachment relationships (such as that with mother) may have more wide-ranging effects. Unfortunately, these points are difficult to investigate and have not been the subject of much discussion in the theoretical literature, despite their centrality. In consequence, the process of internalization, the number and form of internal models, and the way these models influence future behavior are all inadequately understood issues.

Attachment Status and Other Childhood Characteristics. Many studies have examined the relationship between attachment status and other concurrent infant characteristics. Great caution must be exercised in inferring causality on the basis of such evidence. Other studies have looked for relations between attachment status and later social or cognitive characteristics. These studies are more relevant to the hypothesis that early attachments become internalized and shape later development. Evidence emerging from prospective designs cannot be accepted uncritically, however. Attachment status and later developmental characteristics may both be products of a common third factor, such as maternal sensitivity or the quality of the home environment. If this is the case, the temporal order of measurement may provide only the illusion of causality. Nevertheless, this type of evidence is currently the best available.

A variety of studies have assessed the cognitive correlates of secure attachment. No differences in cognitive development across attachment categories are typically found using the Bayley Developmental Scales (Matas, Arend, & Sroufe, 1978; Pastor, 1981), although Main (1983) found that securely attached infants had higher Developmental Quotients than insecure infants of the same age. Frodi, Bridges, and Grolnick (1985) found that at 12 months ambivalent babies demon-

strated less persistence on mastery-related play tasks than secure or avoidant babies, and that avoidant babies demonstrated less overall competence than secure or ambivalent babies. Neither of these differences were sustained at 20 months. Matas, Arend, and Sroufe (1978) observed that securely attached infants engaged in more symbolic play and were more enthusiastic in a problem-solving task than insecure children. Studies have similarly indicated that securely attached infants exhibit more sophisticated spontaneous play than avoidant or ambivalent infants (Belsky, Garduque, & Hrncir, 1984; Slade, 1987). Others find that securely attached infants perform better on an object permanence task than insecure groups (Ahmad & Worobey, 1984; Bell, 1970), though Levitt, Antonucci, and Clark (1984) found no group differences in awareness of object or person permanence. Finally, secure infants have been found to recognize themselves in a mirror both more readily (Schneider-Rosen & Cicchetti, 1984) and less readily (Lewis, Brooks-Gunn, & Jaskir, 1985) than insecure infants.

More directly related to the predictions of attachment theory than general cognitive development is competence at exploring the environment, since the provision of a secure base by the attachment figure should promote exploration. Cassidy (1986) found that secure infants made significantly fewer errors in negotiating their environment during the Strange Situation than avoidant or ambivalent infants, though deriving both attachment status and the competence measure from the same videotapes makes interpretation difficult. Hazen and Durrett (1982) examined exploration and cognitive mapping abilities in 28 children (who had been classified at 12 months) at play in the laboratory when they were between 30 and 34 months. Although the idiosyncratic grouping of infants according to sub-categories of attachment makes comparisons with other studies difficult, they found that secure infants were favored on five of eight measures; however, on only two (one each of exploration and cognitive mapping) was this group superior to both anxiously attached groups. Overall, the evidence suggests that secure infants have a greater propensity to explore their environment than their insecure counterparts.

From a clinical perspective, the most important predictions of attachment theory concern ego functioning and social relations, rather than cognitive abilities. Securely attached infants should be more confident and show more adaptive and harmonious relationships with others. Many studies have examined this area and to a great extent the contention is supported. One paradigm examines the infant's response to adult strangers. Lamb et al. (1982) found that sociability toward a stranger was highest among infants securely attached to father, but was unrelated to attachment status with mother. Main and Weston (1981) reported that secure attachments both with mother and with father predicted more positive interactions with a clown at 12 months, but that the relationship to mother was the strongest predictor. In a prospective study, Lutkenhaus, Grossmann, and Grossmann (1985) assessed attachment to mother at 12 months and interaction with an adult visitor to the home at 3 years. Readiness to interact with the visitor was lower among avoidant children than the other two groups, which did not differ.

Other studies have examined interactions with peers. Jacobson and Wille (1986) observed that 18-month attachment status to mother did not predict sociability or responsiveness to a securely attached peer at 2 or 3 years, but did relate to behavior directed toward the focal children by the peer. At age 3 securely attached

children received more positive responses than insecure children. Avoidant children received fewer positive initiations than others, while ambivalent children received more disruptive responses, more combative initiations, and more resistance to their own combativeness. Patterns at age two were generally weaker. Pastor (1981) found that 20- to 23-month-old children who earlier had been classified as securely attached scored higher on measures of sociability and orientation toward a peer than did members of the two insecure groups. Secure children also redirected their attention after a dispute over a plaything more often than avoidant children. Finally, avoidant and secure children made more social bids than ambivalent children, and ignored fewer social bids made by their playmates.

Several studies have evaluated social characteristics more generally. Main (1983) found secure infants assessed at 12 months to be more sociable and playful in interactive play with an adult woman at 21 months, although the differences in cognitive development noted earlier may confound these findings. Waters, Wippman, and Sroufe (1979) classified infants at 15 months, then observed the children at preschool over a 5-week period at 3.5 years. Five of 12 items regarding confidence and effectiveness favored the secure group, as did 11 of 12 measures of the childrens' competence in interactions with peers.

Erickson, Sroufe, and Egeland (1985; see also Sroufe, Fox, & Pancake, 1983) assessed a sample of 96 children in the Strange Situation at 12 and 18 months, then in a variety of preschool settings (including a special laboratory preschool for 40 of the children) at 4.5 to 5 years. Secure children were rated higher than ambivalent children on a measure of confidence/assertiveness, and avoidant children were rated higher than secure children on dependency. No differences were found on measures of ego control, social skills, positive or negative affect, or compliance, although a more restricted study of the laboratory preschool sample found secure infants to have more positive affect, better social skills, more empathy, and less dependency than anxiously attached infants (Sroufe et al., 1983). On measures completed by teachers, avoidant infants were rated more withdrawn, exhibitionistic and impulsive than secure infants, and gave up at tasks and cried more easily. Avoidant infants in the subsample were described as more hostile and isolated, while ambivalent infants were described as helpless and having low ego control. In contrast, Bates, Maslin, and Frankel (1985) found no relationship between early attachment status and subsequent behavioral difficulties in a study of 3-year-olds who had undergone the Strange Situation at 13 months. They attribute this to their use of a more stable, supportive sample of parents and to having fewer insecurely attached children than in other studies. Their questionnaire method of assessment may also have been less sensitive than the use of trained observers in situ.

In a study by Lewis, Feiring, McGuffog, and Jaskir (1984), avoidant 6-year-old boys (assessed at 12 months with an abbreviated Strange Situation) were rated more schizoid on the Child Behavior Profile (CBP) than secure boys, while ambivalent boys ranked higher on depression and social withdrawal. Both insecure groups showed more uncommunicative behavior and somatic complaints than secure boys. There were no group differences for girls. Using CBP scores it was determined that risk for pathology was 40% for avoidant and ambivalent males compared to 6% of secure males, while girls again showed no differences.

Lewis et al. interpreted these figures to indicate that while early insecure attachment is not sufficient in itself to produce pathology, these boys may be more subject to difficulty when placed under environmental stress.

In summary, while existing evidence is somewhat ambiguous regarding the relationship of attachment status to cognitive faculties, attachment may be relevant to some narrowly defined areas, particularly those involving motivation and confidence in cognitive tasks. In keeping with the central principle that attachment figures serve to enable and encourage exploration, securely attached infants appear to be more accomplished explorers than their insecure counterparts. Secure infants also seem to be more amenable to social contacts, and as older children are more competent at social interaction than their insecure counterparts. They seem to be less easily threatened, and may exhibit fewer behavior disorders than avoidant or ambivalent children.

These findings emerge from studies examining functioning both concurrent with Strange Situation classification and subsequent to it, up to 6 years. Observations of these children into the school years and beyond will be very telling. Findings that early attachment status relates to later adaptation, even as the direct influence of parents is in decline, would be far more compelling evidence of attachment's causal role than is currently available.

The Strange Situation has proven to be an important and useful research tool in the examination of the relationship between the infant and its caretakers. Unfortunately it is sharply limited, to an extent matched by few psychological instruments in wide use, in the age range of subjects to which it can be applied. This makes the assessment of shifting patterns of attachment over time very difficult. If progress is to be made beyond examining the dynamics of attachment in early childhood, then other reliable measures with greater flexibility and for use with older populations must be developed. Promising directions include the recent development of a Q-sort measure of security of attachment (Waters & Deane, 1985) and of a classification procedure that can be applied to 6-year-olds (Main, Kaplan, & Cassidy, 1985). Until such approaches are applied more widely, clinical histories and imperfect subjective measures will limit insight into the influence of attachment on later life.

Despite limitations on the assessment of attachment beyond infancy, there remain questions which must be asked of the theory as it applies to later childhood and adulthood. The final issue to be addressed in the present section is the evidence from studies concerning the internalization of the early attachment relationship and its influence on later functioning.

ATTACHMENT INFLUENCES BEYOND EARLY CHILDHOOD

The research examining Strange Situation behavior and social functioning in infants and children is consistent with the suggestion that early attachment relationships contribute to the individual's representation of others. But, as noted above, problems of assigning causality limit our conclusions. Proponents of this aspect of attachment theory must rely on convergent evidence from multiple perspectives. Although limited, evidence is available to suggest a link between early attachment experience and later parenting, romantic relationships, and depression.

Attachment and Subsequent Parenting

Several studies have examined the relationship between women's parenting behavior (and, in some cases, the attachment status of their children) and their recollections of their own relationships with their mothers. Central to this approach is the often-replicated finding that abusive mothers were frequently themselves abused as children (Belsky, 1978, 1980; DeLozier, 1982; Spinetta & Rigler, 1972). At first glance, a simple modeling process seems the most parsimonious explanation of these observations. Attachment theory does not provide an obvious account of how early aversive interaction would cause individuals to re-enact the abusive actions of their own parents. However, Sroufe and Fleeson (1986) argue that in any relationship the individual learns not only his or her own role, but also the role of the other person: both sides of the relationship are introjected. Thus, the abused child comes to understand relationships as involving an abuser and an abused, and structures (or understands) subsequent relationships in a like manner. The role played may depend on the situation. In parenthood, the role of the abused is occupied by the infant and so the role of abuser is adopted. Role selection may not always be guided by context, however. Bowlby (1973) and Bacciagaluppi (1985) describe situations in which mothers with overwhelming dependency needs attempt to use their children as attachment figures, effectively taking the child role and thrusting the parent role on their children.

Main (Main, Kaplan, & Cassidy, 1985; Main & Goldwyn, 1984) has developed the Adult Attachment Interview as a tool to investigate attachment issues in adulthood. This structured interview involves both a retrospective examination of early attachments and an assessment of current attitudes toward attachment. Initial research has revealed a striking relation between classifications of parents based on this instrument and the Strange Situation categorization of their children. Parents of securely attached children (at 12 and 18 months, and 6 years of age) were comfortable with and valued attachments in adulthood. Recollections of relationships with their own parents were not uniformly positive, but were realistic and not idealized. Importantly, the representation of the relationship, rather than specific early experiences, appeared to be the factor that distinguished them from parents of less securely attached children.

Parents rated as insecure fell into three categories. One group was very dismissive of the importance of attachments and tended to be parents of insecure-avoidant children (A). A second group was preoccupied with and often idealized their relationships with their own parents. These parents most frequently had children categorized as insecure–ambivalent (C). A final group of parents had experienced the death of an attachment figure in childhood and had not successfully resolved this loss. Such individuals were most frequently parents of insecure–disorganized/disoriented infants (D). Overall, Main et al. (1985) reported a significant correlation ($r=.62$) between infant classification and the mother's working model of attachment as assessed using the Adult Attachment Interview. These results provide perhaps the first empirical validation of the influential hypothesis that the parent's representations of past attachments influence behavior with their own children and, ultimately, the form of the new parent–child relationship.

Ricks (1985) found that mothers of secure infants rated their own mothers significantly higher on acceptance and encouragement of their independence than mothers of insecure infants, and significantly lower on overprotection and rejec-

tion. Aggregate ratings of maternal encouragement and acceptance were also higher for mothers of secure infants. In addition, these mothers rated their fathers more favorably on encouragement of independence, acceptance, and both aggregate scales, but showed no differences on overprotection or rejection. No differences on rated idealization of the mother's mother were found, but mothers of secure infants tended to idealize their fathers more than mothers of insecure infants. In a followup study (1985), mothers of insecure infants were found to be more defensive and idealized *both* parents more than mothers of secure infants.

Sroufe and Fleeson (1986) describe a study by Morris (1980) in which 9 of 36 mothers exhibited seductive behavior toward their infants, defined as fondling, sensual teasing, kissing on the lips, and so on. This behavior was directed almost entirely toward sons. Of the nine mothers, eight were identified as having been the subject of emotional manipulation or incest by their fathers. Only 4 of 12 such victims failed to act in a seductive manner toward their children; of these, three were observed only with daughters. Sroufe and Fleeson argue that modelling does not appear to be an adequate explanation for these findings, as the sex of child and the specific behaviors engaged in by the mothers and their fathers differed. Instead, these women imitated the sexual nature of their earlier relationships while tailoring the behaviors and selection of subject to the female sex-role. This suggests that it was the pattern of relating, not the behavior, that was learned in the earlier relationship.

Although scarce, the available research suggests that early experience influences adult parenting behavior. The prospective mother seems to introject early relationships and uses these as templates for relationships with her children. This conclusion rests on retrospective data, however, and maternal recollections may be inaccurate. Reports may be biased, perhaps in part by the salience of the present relationship with the infant, creating spurious correlations. Recall bias might be particularly attractive to abusive parents, who might find in such strategies justifications or rationales for their own behavior.

Helpful support for the attachment view comes from studies in which the past relationship variables were externally verifiable or, at least, were reasonably discrete and therefore perhaps less subject to retrospective distortion than ratings of parental style. Frommer and O'Shea (1973a, 1973b) found that mothers who reported that they had been separated from either parent before the age of 11 exhibited difficulties in parenting, marital problems, and depression. Hall and Pawlby (1981) found that women from disrupted families seemed less involved with their babies. Finally, Rutter, Quinton, and Liddle (1983) observed that mothers whose children had been taken into government care on different occasions were more than three times as likely as matched controls to have been in government care in childhood themselves, and were also far more likely to have been separated from their parents due to apparent conflict or rejection.

Attachment and Subsequent Romantic Relationships

The available literature on attachment as it relates to romantic and peer relationships is more limited. Weiss (1982) notes that some relationships between adults (most commonly romantic relationships and relationships in situations of extreme stress) seem to satisfy the requirements of attachment bonds: the individual is motivated to remain within range of the attachment figure, the presence of

the figure is accompanied by increased comfort and security, and separations or threats to access to the figure may be perceived as threats to the individual's own well-being and may be met with protest. On the other hand, Weiss observes, adult relationships differ from childhood attachment in at least three ways: these relationships are with peers, not older and wiser caretakers; threats to continued access do not overwhelm other behavior as easily as in infants; and in the context of romantic relationships there is the added explicit component of sexuality. In a subsequent article Weiss (1986) presents a typology of bonds in adulthood and points out that only a few qualify even as approximations of attachment relationships. It should be noted, however, that the internal working models developed in childhood may influence even relationships in which attachment behavior is not elicited.

Empirical research on this topic is noticeably sparse. Hetherington (1972) found that adolescent girls who had lost a father to divorce sought more attention from an adult male—perhaps demonstrating a dependent, or anxiously attached, style—than did girls without such loss. Girls who had lost a father through death were more avoidant of an adult male than the other two groups. Hepworth, Ryder, and Dreyer (1984) examined courtship patterns among college students who had or had not experienced recent parental loss. They observed a trend among the loss sample toward either accelerated courtship or to increased hesitancy regarding close relationships relative to the control sample. As with Hetherington, those with a loss due to death more often avoided close relationships, while those with a loss due to divorce had accelerated courtship and more interest in relationships. The recency of these losses (generally within the previous 5 years) suggests that the effects might be attributable to residual mourning rather than formative influences on internal working models, although such influences may not end in early childhood.

Attachment and Subsequent Psychopathology

Research has also related depression and suicidal behavior and ideation to early childhood experiences with caretakers. As described above, Frommer and O'Shea (1973b) found that early separation predicted self-reported depression. Lindeman (1960) found that patients with early separation also had high rates of both suicidal ideation and depressive episodes. In a sample of 79 women whose mothers had died while the subjects were children, Parker and Hadzi-Pavlovic (1984) found that subsequent lack of caring by fathers and stepmothers significantly predicted high trait depression in adulthood. Tennant, Hurry, and Bebbington (1982) found that while none of four types of parental separation from birth to age four predicted psychological morbidity in adulthood, several measures of separation at age 5–10 predicted both overall morbidity and depression, as did separation due to parental illness from age 11 to 15.

With university students, Adam (1982) found that parental loss due to death and divorce was strongly related to later self-reported suicidal ideation and suicide attempts. A subsequent study revealed family instability to be a significant factor related to loss, with greater instability in the histories of subjects who had attempted suicide. Adam concludes that the literature on loss in childhood supports a predictive link to later attempted suicide, although the type of loss and age at loss associated with greatest vulnerability remain a matter of debate.

What little evidence exists suggests that early attachment status and subsequent loss influence later psychopathology and behavior in relationships. The studies available are inevitably flawed, however, and there is not the breadth of evidence from different perspectives utilizing different paradigms that would be necessary to offset this shortcoming. Although the evidence is substantially supportive of this critical aspect of attachment theory, the point cannot be said to have been established with the rigour one would prefer as a basis for clinical intervention.

SUMMARY

Attachment theory has spawned a large and ever expanding body of research, most of it concerned with functioning in infancy. A number of the fundamental propositions of the theory have been supported. The available data suggest that human infants have a primary need for social interaction of some kind, perhaps specifically involving an exclusive relationship with a single caretaker perceived as "older and/or wiser." Not supported by the data (and not a part of Bowlby's original theory) is the concept of a critical period soon after birth for the formation of the mother's bond to the child. The attachment status of the child appears to depend, as would be predicted, on the personality and caretaking style of the mother, which typically seem to remain constant in the absence of unusual stresses. The infant's attachment pattern may differ with different caretakers.

Caretaker variables associated with secure attachment in the infant are psychological health, responsiveness to the infant's attachment cues, and the absence of abuse or neglect. Secure attachment appears to promote certain limited aspects of cognitive development and more wide-ranging aspects of social development, with securely attached infants being more sociable and eliciting more positive responses from others. Finally, early attachment relationships may contribute to a more adaptive model of the self in relation to others, both in caretaking situations and close adult relationships, though few studies have investigated the latter. In addition to gaps in the research literature, a major difficulty is a lack of consensus regarding the issue of how and when internal working models are formed and altered.

The lack of direct evidence in support of the influence of early attachment on later interpersonal functioning is not accompanied by a body of research searching for such evidence and failing to find it. The difficulty lies in formulating and carrying out practical research designs that can adequately test the theory. Nor can it be said that the theory is untestable by its very nature. Practical and ethical concerns, however, limit the types of study which can be undertaken. This represents one of the many instances in clinical practice in which we may be forced to content ourselves with indirect evidence from a number of different sources, perspectives, and paradigms. Where this is available the tenets of attachment theory are supported to a degree. We may never have direct evidence that internal working models developed in early life form the basis of adult relationships. Instead we may have to assess attachment theory from a practical perspective, determining with each client whether it provides a useful model of adult functioning.

Acknowledgements — Preparation of this manuscript was supported by a Medical Research Council of Canada Doctoral Studentship to the first author, Natural Sciences and Engi-

neering Research Council of Canada Grant A7046 to the second author, and Medical Research Council of Canada Operating Grant MA5028 to Richard W. J. Neufeld.

The authors thank Joel Herscovitch and Richard W. J. Neufeld for their helpful comments on an earlier draft of this article.

REFERENCES

Adam, K. S. (1982). Loss, suicide, and attachment. In C. M. Parkes & J. Stevenson-Hinde (Eds.), *The place of attachment in human behavior* (pp. 269–294). New York: Basic Books.

Ahmad, A., & Worobey, J. (1984). Attachment and cognition in a naturalistic context. *Child Study Journal*, **14**, 185–203.

Ainsworth, M. D. S. (1982). Attachment: Retrospect and prospect. In C. M. Parkes & J. Stevenson-Hinde (Eds.), *The place of attachment in human behavior* (pp. 3–30). New York: Basic Books.

Ainsworth, M. D. S., Blehar, M. C., Waters, E., & Wall, S. (1978). *Patterns of attachment: A psychological study of the Strange Situation*. Hillsdale, NJ: Earlbaum.

Ainsworth, M. D. S., & Wittig, B. A. (1969). Attachment and exploratory behavior of one year olds in a strange situation. In B. M. Foss (Ed.), *Determinants of infant behavior* (Vol. 4, pp. 111–136). London: Methuen.

American Psychiatric Association. (1987). *Diagnostic and statistical manual of mental disorders* (3rd ed., revised). Washington, DC: Author.

Bacciagaluppi, M. (1985). Inversion of parent-child relationships: A contribution to attachment theory. *British Journal of Medical Psychology*, **58**, 369–373.

Bates, J. E., Maslin, C. A., & Frankel, K. A. (1985). Attachment security, mother-child interaction, and temperament as predictors of behavior-problem ratings at age three years. *Monographs of the Society for Research in Child Development*, **50**, 167–193.

Bell, S. M. (1970). The development of the concept of object as related to infant-mother attachment. *Child Development*, **41**, 291–311.

Belsky, J. (1978). Three theoretical models of child abuse: A critical review. *Child Abuse and Neglect*, **2**, 37–49.

Belsky, J. (1980). Child maltreatment: An ecological integration. *American Psychologist*, **35**, 320–335.

Belsky, J., Garduque, L., & Hrncir, E. (1984). Assessing performance, competence, and executive capacity in infant play: Relations to home environment and security of attachment. *Developmental Psychology*, **20**, 406–417.

Belsky, J., Rovine, M., & Taylor, D. G. (1984). The Pennsylvania Infant and Family Development Project, III: The origins of individual differences in infant-mother attachment: Maternal and infant contributions. *Child Development*, **55**, 718–728.

Benn, R. K. (1986). Factors promoting secure attachment relationships between employed mothers and their sons. *Child Development*, **57**, 1224–1231.

Bowlby, J. (1957). An ethological approach to research in child development. *British Journal of Medical Psychology*, **30**, 230–240.

Bowlby, J. (1958). Psychoanalysis and child care. In J. D. Sutherland (Ed.), *Psychoanalysis and contemporary thought* (pp. 33–57). London: Hogarth Press.

Bowlby, J. (1961). Childhood mourning and its implications for psychiatry. *American Journal of Psychiatry*, **118**, 481–498.

Bowlby, J. (1969). *Attachment and loss: Vol. 1. Attachment*. New York: Basic Books.

Bowlby, J. (1973). *Attachment and loss. Vol. 2. Separation: Anxiety and Anger*. New York: Basic Books.

Bowlby, J. (1977a). The making and breaking of affectional bonds. I. Aetiology and psychopathology in the light of attachment theory. *British Journal of Psychiatry*, **130**, 201–210.

Bowlby, J. (1977b). The making and breaking of affectional bonds. II. Some principles of psychotherapy. *British Journal of Psychiatry*, **130**, 421–431.

Bowlby, J. (1980). *Attachment and loss: Vol. 3. Loss, sadness, and depression*. New York: Basic Books.

Bowlby, J. (1984). *Attachment and loss: Vol. 1. Attachment* (rev. ed.). New York: Basic Books.

Bowlby, J., & Parkes, C. M. (1970). Separation and loss within the family. In E. J. Anthony & C. Koupernik (Eds.), *The child in his family* (Vol. 1, pp. 197–216). New York: John Wiley & Sons.

Bretherton, I. (1985). Attachment theory: Retrospect and prospect. *Monographs of the Society for Research in Child Development*, **50**, 3–35.

Call, J. D. (1984). Child abuse and neglect in infancy: Sources of hostility within the parent–infant dyad and disorders of attachment in infancy. *Child Abuse and Neglect*, **8**, 185–202.

Cassidy, J. (1986). The ability to negotiate the environment: An aspect of infant competence as related to quality of attachment. *Child Development*, **57**, 331–337.

Clark, E. A., & Hanisee, J. (1982). Intellectual and adaptive performance of Asian children in adoptive American settings. *Developmental Psychology*, **18**, 595–599.

Coates, B., Anderson, E. P., & Hartup, W. W. (1972). Interrelations in the attachment behavior of human infants. *Developmental Psychology*, **6**, 218–230.

Cottrell, L. S., Jr. (1969). Interpersonal interaction and the development of the self. In D. A. Goslin (Ed.), *Handbook of socialization theory and research* (pp. 543–570). Chicago: Rand-McNally.

D'Angelo, E. J. (1986). Security of attachment in infants with schizophrenic, depressed, and unaffected mothers. *Journal of Genetic Psychology*, **147**, 421–422.

DeLozier, P. (1982). Attachment theory and child abuse. In C. Parkes & J. Stevenson-Hinde (Eds.), *The place of attachment in human behavior* (pp. 95–117). New York: Basic Books.

Dennis, W. (1973). *Children of the creche*. New York: Appleton-Century-Crofts.

Egeland, B., & Farber, E. A. (1984). Infant–mother attachment: Factors related to its development and changes over time. *Child Development*, **55**, 753–771.

Egeland, B., & Sruofe, L. A. (1981a). Attachment and early maltreatment. *Child Development*, **52**, 44–52.

Egeland, B., & Sroufe, L. A. (1981b). Developmental sequelae of maltreatment in infancy. In R. Rizley & D. Cicchetti (Eds.), *Developmental perspectives on child maltreatment* (pp. 77–92). San Francisco: Jossey-Bass.

Egeland, B., & Vaughn, B. (1981). Failure of "bond formation" as a cause of abuse, neglect, and maltreatment. *American Journal of Orthopsychiatry*, **51**, 78–84.

Erickson, M. F., Sroufe, L. A., & Egeland, B. (1985). The relationship between quality of attachment and behavior problems in preschool in a high-risk sample. *Monographs of the Society for Research in Child Development*, **50**, 147–166.

Frodi, A., Bridges, L., & Grolnick, W. (1985). Correlates of mastery-related behavior: A short-term longitudinal study of infants in their second year. *Child Development*, **56**, 1291–1298.

Frommer, E. A., & O'Shea, G. (1973a). Antenatal identification of women liable to have problems in managing their infants. *British Journal of Psychiatry*, **123**, 149–156.

Frommer, E. A., & O'Shea, G. (1973b). The importance of childhood experience in relation to problems of marriage and family building. *British Journal of Psychiatry*, **123**, 157–160.

Gaensbauer, T. J., Harmon, R. J., Culp, A. M., Schultz, L. A., van Doorninck, W. J., & Dawson, P. (1985). Relationships between attachment behavior in the laboratory and the caretaking environment. *Infant Behavior and Development*, **8**, 355–369.

Gomes-Pedro, J., de Almeida, J. B., da Costa, C. S., & Barbosa, A. (1984). Influence of early mother-infant contact on dyadic behaviour during the first month of life. *Developmental Medicine and Child Neurology*, **26**, 657–664.

Grossmann, K., & Grossmann, K. (1981). Parent–infant attachment relationships in Bielefeld. In K. Immelmann, G. W. Barlow, L. Petrinovich, & M. Main (Eds.), *Behavioral development: The Bielefeld interdisciplinary project* (pp. 694–699). New York: Cambridge University Press.

Grossmann, K. E., Grossmann, K., Huber, F., & Wartner, U. (1981). German children's behavior towards their mothers at 12 months and their fathers at 18 months in Ainsworth's Strange Situation. *International Journal of Behavioral Development*, **4**, 157–181.

Hall, F., & Pawlby, S. (1981). Continuity and discontinuity in the behavior of British working-class mothers and their first-born children. *International Journal of Behavioral Development*, **4**, 13–26.

Harlow, H. F. (1958). The nature of love. *American Psychologist*, **13**, 673–685.

Harlow, H. F. (1971). *Learning to love*. San Francisco: Albion.

Hazen, N. L., & Durrett, M. E. (1982). Relationship of security of attachment to exploration and cognitive mapping abilities in 2-year-olds. *Developmental Psychology*, **18**, 751–759.

Hepworth, J., Ryder, R. G., & Dreyer, A. S. (1984). The effects of parental loss on the formation of intimate relationships. *Journal of Marital and Family Therapy*, **10**, 73–82.

Hetherington, E. M. (1972). Effects of father absence on personality development in adolescent daughters. *Developmental Psychology*, **7**, 313–326.

Jacobson, J. L., & Wille, D. E. (1986). The influence of attachment pattern on developmental changes in peer interaction from the toddler to the preschool period. *Child Development*, **57**, 338–347.

Jensen, G. D. (1965). Mother–infant relationship in the monkey *Macaca nemestrina*: Development of specificity of maternal response to own infant. *Journal of Comparative and Physiological Psychology*, **59**, 305–308.

Kagan, J. (1982). *Psychological research on the human infant: An evaluative summary*. New York: W. T. Grant Foundation.

Kennell, J. H., Jerauld, R., Wolfe, H., Chesler, D., Kreger, N. C., McAlpine, W., Steffa, M., & Klaus, M. H. (1974). Maternal behavior one year after early and extended post-partum contact. *Developmental Medicine and Child Neurology*, **16**, 172–179.

Klaus, M. H., Jerauld, R., Kreger, N. C., McAlpine, W., Steffa, M., & Kennell, J. H. (1972). Maternal attachment: Importance of the first post-partum days. *New England Journal of Medicine*, **286**, 460–463.

Klaus, M. H., & Kennell, J. H. (1976). *Maternal-infant bonding*. St. Louis, MO: Mosby.

Klaus, M. H., & Kennell, J. H. (1982). *Parent-infant bonding*. St. Louis, MO: Mosby.

Lamb, M. E. (1974). A defense of the concept of attachment. *Human Development*, **17**, 376–385.

Lamb, M. E. (1977). Father–infant and mother–infant interaction in the first year of life. *Child Development*, **48**, 167–181.

Lamb, M. E. (1978). Qualitative aspects of mother– and father–infant attachments. *Infant Behavior and Development*, **1**, 265–275.

Lamb, M. E. (1982). Early contact and maternal-infant bonding: One decade later. *Pediatrics*, **70**, 763–768.

Lamb, M. E., Gaensbauer, T. J., Malkin, C. M., & Schultz, L. A. (1985). The effects of child maltreatment on security of infant–adult attachment. *Infant Behavior and Development*, **8**, 35–45.

Lamb, M. E., Hwang, C. P., Frodi, A. M., & Frodi, M. (1982). Security of mother- and father-infant attachment and its relation to sociability with strangers in traditional and nontraditional Swedish families. *Infant Behavior and Development*, **5**, 355–367.

Lamb, M. E., Thompson, R. A., Gardner, W., & Charnov, E. L. (1985). *Infant-mother attachment: The origins and developmental significance of individual differences in strange situation behavior*. Hillsdale, NJ: Erlbaum.

Lamb, M. E., Thompson, R. A., Gardner, W. P., Charnov, E. L., & Estes, D. (1984). Security of infantile attachment as assessed in the Strange Situation: Its study and biological interpretation. *Behavioral and Brain Sciences*, **7**, 127–147.

Levitt, M. J., Antonucci, T. C., & Clark, M. C. (1984). Object–person permanence and attachment: Another look. *Merrill-Palmer Quarterly*, **30**, 1–10.

Levy-Shiff, R. (1983). Adaptation and competence in early childhood: Communally-raised kibbutz children versus family raised children in the city. *Child Development*, **54**, 1606–1614.

Lewis, M., Brooks-Gunn, J., & Jaskir, J. (1985). Individual differences in visual self-recognition as a function of mother–infant attachment relationship. *Developmental Psychology*, **21**, 1181–1187.

Lewis, M., Feiring, C., McGuffog, C., & Jaskir, J. (1984). Predicting psychopathology in six-year-olds from early social relations. *Child Development*, **55**, 123–136.

Lindeman, E. (1960). Psychosocial factors as stress agents. In J. M. Tanner (Ed.), *Stress and psychiatric disorder* (pp. 112–137). Oxford: Blackwell.

Lorenz, K. (1952). *King Solomon's ring*. London: Methuen.

Lutkenhaus, P., Grossmann, K. E., & Grossmann, K. (1985). Infant–mother attachment at twelve months and style of interaction with a stranger at the age of three years. *Child Development*, **56**, 1538–1542.

Main, M. (1983). Exploration, play, and cognitive functioning related to infant–mother attachment. *Infant Behavior and Development*, **6**, 167–174.

Main, M., & Goldwyn, R. (1984). Predicting rejection of her infant from mother's representation of her own experience: Implications for the abused-abusing intergenerational cycle. *Child Abuse and Neglect*, **8**, 203–217.

Main, M., Kaplan, N., & Cassidy, J. (1985). Security in infancy, childhood, and adulthood: A move to the level of representation. *Monographs of the Society for Research in Child Development*, **50**, 66–104.

Main, M. B., & Weston, D. R. (1981). The quality of the toddler's relationship to mother and to father: Related to conflict behavior and the readiness to establish new relationships. *Child Development*, **52**, 932–940.

Marvin, R. S. (1977). An ethological-cognitive model for the attenuation of mother–child attachment behavior. In T. Alloway, P. Pliner, & L. Krames (Eds.), *Advances in the study of communication and affect: Vol. 3. Attachment behavior* (pp. 25–60). New York: Plenum Press.

Matas, L., Arend, R. A., & Sroufe, L. A. (1978). Continuity of adaptation in the second year: The relationship between quality of attachment and later competence. *Child Development*, **49**, 547–556.

Morris, D. (1980). *Infant attachment and problem solving in the toddler: Relations to mother's family history.* Unpublished doctoral dissertation, University of Minnesota.

Myers, B. J. (1984). Mother–infant bonding: The status of this critical-period hypothesis. *Developmental Review*, **4**, 240–274.

Naslund, B., Persson–Blennow, I., McNeil, T., Kaij, L., & Malmquist-Larsson, A. (1984). Offspring of women with nonorganic psychosis: Infant attachment to the mother at one year of age. *Acta Psychiatrica Scandinavia*, **69**, 231–241.

Owen, M. T., & Chase-Lansdale, L. (1982, April). *Similarity between infant-mother and infant-father attachments.* Paper presented at the meeting of the Southwestern Society for Research in Human Development, Galveston, TX.

Parker, G. & Hadzi-Pavlovic, D. (1984). Modification of levels of depression in mother-bereaved women by parental and marital relationships. *Psychological Medicine*, **14**, 125–135.

Parkes, C. M. (1972). *Bereavement: Studies of grief in adult life.* New York: International Universities Press.

Parkes, C. M. (1982). Attachment and the prevention of mental disorders. In C. M. Parkes & J. Stevenson-Hinde (Eds.), *The place of attachment in human behavior* (pp. 295–309). New York: Basic Books.

Parkes, C. M. (1985). Bereavement. *British Journal of Psychiatry*, **146**, 11–17.

Pastor, D. L. (1981). The quality of mother–infant attachment and its relationship to toddlers' initial sociability with peers. *Developmental Psychology*, **17**, 326–335.

Pollitt, E., Eichler, A. W., & Chan, C. K. (1975). Psychosocial development and behavior of mothers of failure-to-thrive children. *American Journal of Orthopsychiatry*, **45**, 525–537.

Provence, S., & Lipton, R. C. (1962). *Infants in institutions.* New York: International Universities Press.

Radke-Yarrow, M., Cummings, E. M., Kuczynski, L., & Chapman, M. (1985). Patterns of attachment in two- and three-year-olds in normal families and families with parental depression. *Child Development*, **56**, 884–893.

Raphael, B. (1982). The young child and the death of a parent. In C. M. Parkes & J. Stevenson-Hinde (Eds.), *The place of attachment in human behavior* (pp. 131–150). New York: Basic Books.

Raphael, B. (1983). *The anatomy of bereavement.* New York: Basic Books.

Ricks, M. H. (1985). The social transmission of parental behavior: Attachment across generations. *Monographs of the Society for Research in Child Development*, **50**, 211–227.

Ringler, N., Kennell, J. H., Jarvella, R., Navojosky, B. J., & Klaus, M. H. (1975). Mother-to-child speech at 2 years: Effects of early postnatal contact. *Journal of Pediatrics*, **86**, 141–144.

Ringler, N., Trause, M. A., Klaus, M., & Kennell, J. (1978). The effects of extra postpartum contact and maternal speech patterns on children's IQs, speech, and language comprehension at five. *Child Development*, **49**, 862–865.

Rogers, C. (1951). *Client-centered therapy.* Boston: Houghton Mifflin.

Rutter, M. (1979). Maternal deprivation, 1972–1978: New findings, new concepts, new approaches. *Child Development*, **50**, 283–305.

Rutter, M., Quinton, D., & Liddle, C. (1983). Parenting in two generations: Looking backwards and looking forwards. In N. Madge (Ed.), *Families at risk* (pp. 60–98). London: Heineman.

Sagi, A., Lamb, M. E., Lewkowicz, K. S., Shoham, R., Dvir, R., & Estes, D. (1985). Security of infant–mother, father, and metapelet attachments among kibbutz-reared Israeli children. *Monographs of the Society for Research in Child Development*, **50**, 257–275.

Schneider-Rosen, K., Braunwald, K. G., Carlson, V., & Cicchetti, D. (1985). Current perspectives in attachment theory: Illustration from the study of maltreated infants. *Monographs of the Society for Research in Child Development*, **50**, 194–210.

Schneider-Rosen, K., & Cicchetti, D. (1984). The relationship between affect and cognition in maltreated infants: Quality of attachment and the development of visual self-recognition. *Child Development*, **55**, 648–658.

Slade, A. (1987). Quality of attachment and early symbolic play. *Developmental Psychology*, **23**, 78–85.

Spinetta, J. J., & Rigler, D. (1972). The child-abusing parent: A psychological review. *Psychological Bulletin*, **77**, 296–304.

Spitz, R. A. (1945). Hospitalism: An inquiry into the genesis of psychiatric conditions in early childhood. *Psychoanalytic Study of the Child*, **1**, 53–74.

Sroufe, L. A. (1985). Attachment classification from the perspective of infant-caregiver relationships and infant temperament. *Child Development*, **56**, 1–14.

Sroufe, L. A., & Fleeson, J. (1986). Attachment and the construction of relationships. In W. W. Hartup & Z. Rubin (Eds.), *Relationships and development* (pp. 51–71). Hillsdale, NJ: Erlbaum.

Sroufe, L. A., Fox, N. E., & Pancake, V. R. (1983). Attachment and dependency in developmental perspective. *Child Development*, **54**, 1615–1627.

Stephen, T. D. (1984). Symbolic interdependence and post-break-up distress: A reformulation of the attachment construct. *Journal of Divorce*, **8**, 1–16.

Svejda, M. J., Campos, J. J., & Emde, R. N. (1980). Mother–infant "bonding": Failure to generalize. *Child Development*, **51**, 775–779.

Tennant, C., Hurry, J., & Bebbington, P. (1982). The relation of childhood separation experiences to adult depressive and anxiety states. *British Journal of Psychiatry*, **141**, 475–482.

Thompson, R. A., Lamb, M. E., & Estes, D. (1982). Stability of infant-mother attachment and its relationship to changing life circumstances in an unselected middle class sample. *Child Development*, **53**, 144–148.

Tibbits-Kleber, A. L., & Howell, R. J. (1985). Reactive attachment disorder of infancy (RAD). *Journal of Clinical Child Psychology*, **14**, 304–310.

Tronick, E. Z., Ricks, M., & Cohn, J. F. (1982). Maternal and infant affective exchange: Patterns of adaptation. In T. Field & A. Fogel (Eds.), *Emotion and early interaction* (pp. 83–100). Hillsdale, NJ: Erlbaum.

Turco, R. (1982). The influence of previous life events and developmental experiences on the resolution of grief as developed in psychotherapy. *Psychiatric Quarterly*, **54**, 143–148.

Vaughn, B., Egeland, B., Sroufe, L. A., & Waters, E. (1979). Individual differences in infant-mother attachment at twelve and eighteen months: Stability and change in families under stress. *Child Development*, **50**, 971–975.

Waters, E., & Deane, K. E. (1985). Defining and assessing individual differences in attachment relationships: Q-methodology and the organization of behavior in infancy and early childhood. *Monographs of the Society for Research in Child Development*, **50**, 41–65.

Waters, E., Wippman, J., & Sroufe, L. A. (1979). Attachment, positive affect, and competence in the peer group: Two studies in construct validation. *Child Development*, **50**, 821–829.

Weiss, R. S. (1982). Attachment in adult life. In C. M. Parkes & J. Stevenson-Hinde (Ed.), *The place of attachment in human behavior* (pp. 171–184). New York: Basic Books.

Weiss, R. S. (1986). Continuities and transformations in social relationships from childhood to adulthood. In W. W. Hartup & Z. Rubin (Eds.), *Relationships and development* (pp. 95–110). Hillsdale, NJ: Erlbaum.